

**Roachbot**

The squish-proof cockroach inspired robot put together by the engineering team at Berkeley University. <https://digit.in/jul20-72>

**HAMR**

The HAMR platform developed by researchers at Harvard is another hardy robot inspired by the cockroach. <https://digit.in/jul20-73>



ROACH TECH

What we can learn from cockroaches

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ost people are disgusted by cockroaches, and rightfully so, since they thrive in dirty,

unhygienic places and spread a number of diseases including cholera, typhoid and dysentery. These are only the ones that humans are

exposed to though, and there are a wide variety of cockroaches that live all around the world without being harmful to humans. Cockroaches are however extremely hardy, are notoriously hard to kill and can survive almost anything that you throw at them, including nuclear weapons. These creatures are at the peak of evolution in many ways, and there are a lot of lessons that scientists are learning from them for human applications.

COCKROACH MILK

The Pacific Beetle Cockroaches give birth to live young, with their birth

canals leading to a chamber in their stomach. Insects do not have placental sacs, so the food exchange with developing embryos is directly through the stomach. *Diploptera punctate* is the only known species of cockroach to give birth to live young. When a university researcher in Iowa named Nathan Coussens was studying these cockroaches, he found that shiny crystals were spilling into their guts. The other scientists dismissed them as organic salts, which are commonly found in the stomachs of insects, but Coussens believed that the substance was something special. Coussens discovered that the cockroaches were actually lactating for their young within the stomach.

The good cockroach mothers release a crystal that is tightly packed with proteins, fats and carbohydrates. In other words, it has all the essential ingredients of a healthy meal, with a high calorie content. The amount of energy derived from this “milk” is more than twice that of Buffalo milk. In simple words, this cockroach milk is better than traditional dairy milks, and has been identified as one of the superfoods of the future, and can be used as an important source of nutrition for the ever increasing global population.

Despite how disgusting it may seem, the milk does not taste like anything in particular, and is pretty bland. It looks like glitter, which makes it more palatable, maybe. It is improbable that we can extract the milk at scale from farms of cockroaches. To produce the milk, researchers from the Institute for Stem Cell Biology and Regenerative Medicine (inStem) in Bengaluru, managed to synthesize the protein milk in 2016. During the synthesis, the researchers discovered that the proteins themselves contained all the essential amino acids required by humans, which could be useful for repairing and maintaining muscle tissue. The researchers from InStem now plan to use yeast to produce the milk in commercial